

WARTH, Austria

WMO: 113080

Lat: 47.256N

Lon: 10.186E

Elev: 4824

StdP: 12.31

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.7	(c) 8.0	(d) -1.1	(e) 6.2	(f) 7.0	(g) 2.1	(h) 7.3	(i) 11.6	(j) 14.5	(k) 35.2	(l) 12.8	(m) 34.2	(n) 2.3	(o) 80	(p) 0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.4	75.2	59.4	72.0	57.9	68.9	56.4	61.5	72.0	59.5	69.1	57.8	66.5	4.4	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.5	84.6	66.0	55.6	78.9	63.1	54.1	74.5	61.4	29.9	72.2	28.4	69.6	27.1	66.5	70.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.0	(b) 9.3	(c) 7.7	(e) -1.4	(f) 79.7	(g) 5.2	(h) 3.1	(i) -5.2	(j) 82.0	(k) -8.2	(l) 83.8	(m) -11.2	(n) 85.6	(o) -14.9	(p) 87.8		
(a) 11.0	(b) 9.3	(c) 7.7	(e) -2.0	(f) 63.4	(g) 5.0	(h) 2.4	(i) -5.6	(j) 65.1	(k) -8.5	(l) 66.5	(m) -11.3	(n) 67.8	(o) -14.9	(p) 69.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.1	26.2	27.6	32.4	38.1	46.0	52.8	55.7	55.6	49.1	44.6	34.7	28.9		
	DBStd	13.09	8.18	9.05	8.19	7.51	6.84	7.78	6.42	6.10	6.56	7.49	9.27	8.74		
	HDD50	3933	738	626	547	359	160	58	24	19	95	192	461	654		
	HDD65	8753	1203	1046	1012	806	589	370	294	295	478	632	909	1119		
	CDD50	668	0	0	0	3	36	144	199	192	67	25	2	0		
	CDD65	14	0	0	0	0	0	5	5	4	0	0	0	0		
	CDH74	136	0	0	0	0	0	43	50	43	0	0	0	0		
	CDH80	8	0	0	0	0	0	5	2	1	0	0	0	0		
	WSAvg	3.1	3.4	3.4	3.5	3.1	3.0	3.0	3.0	2.8	2.8	2.8	2.9	3.0		
	PrecAvg	72.6	4.7	4.7	5.5	4.5	6.7	7.9	8.6	8.3	6.1	4.7	5.3	5.5		
(15) Precipitation	PrecMax	95.1	11.7	14.4	10.5	9.4	16.2	13.3	15.3	16.0	12.1	10.2	13.3	13.6		
	PrecMin	55.4	0.1	0.9	1.8	0.7	1.8	3.3	3.4	2.6	2.2	0.5	0.0	0.5		
	PrecStd	8.1	3.1	2.9	2.6	2.2	2.9	2.4	3.1	3.4	2.2	2.2	2.8	3.0		
	PrecStd	8.1	3.1	2.9	2.6	2.2	2.9	2.4	3.1	3.4	2.2	2.2	2.8	3.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.3	49.7	54.5	61.8	70.6	78.9	78.9	78.3	70.7	66.6	57.6	49.8		
		MCWB	37.7	37.3	41.0	47.9	54.0	59.7	60.3	60.7	56.0	51.2	45.6	38.8		
	2%	DB	43.4	46.1	50.9	58.0	66.1	73.9	74.8	74.6	67.0	62.5	54.0	45.9		
		MCWB	35.4	35.9	39.6	45.3	51.6	58.2	59.4	60.0	54.7	49.0	43.1	36.6		
	5%	DB	40.3	42.6	47.7	54.4	62.1	70.6	71.2	70.8	63.9	59.1	50.8	42.4		
		MCWB	33.9	34.0	38.0	43.2	49.7	57.1	57.9	57.8	53.5	48.0	41.5	35.1		
	10%	DB	37.5	39.4	44.2	50.7	58.3	66.8	67.5	67.3	60.5	55.7	46.9	39.5		
		MCWB	32.4	32.9	36.4	41.2	47.9	55.6	56.7	56.6	51.7	46.7	39.7	34.1		
	0.4%	WB	39.8	39.0	42.5	49.5	56.3	66.1	63.3	63.7	59.1	53.7	47.1	40.3		
		MCDB	44.5	45.9	51.6	59.8	67.4	71.8	73.4	74.1	66.7	61.6	55.9	47.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.8	36.9	40.5	46.6	53.1	60.9	60.9	61.1	56.6	51.1	44.2	38.2		
		MCDB	41.3	43.2	48.5	56.2	62.3	72.1	71.8	71.6	63.9	58.7	51.6	43.4		
	5%	WB	34.9	35.2	38.9	44.0	51.0	58.0	59.1	59.2	54.6	49.4	42.2	36.5		
		MCDB	39.3	41.1	46.2	52.6	59.4	68.1	68.5	68.3	61.7	56.5	49.0	40.9		
	10%	WB	32.9	33.4	37.1	41.9	48.9	56.0	57.4	57.4	52.7	47.8	40.3	34.7		
		MCDB	36.8	38.8	42.9	49.5	56.7	65.1	65.7	65.4	58.9	54.3	46.4	38.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.7	13.9	14.1	14.3	15.1	15.5	15.4	15.4	15.2	15.8	12.9	11.8		
		MCDBR	14.7	17.2	19.3	20.4	22.0	22.1	22.1	22.2	21.2	21.2	17.2	14.9		
		MCWBR	9.7	10.7	10.9	10.7	10.7	10.1	9.9	10.4	11.4	11.0	10.1	9.3		
		MCDBR	13.1	15.1	17.1	18.6	18.8	19.7	19.4	20.4	18.9	18.1	16.0	13.3		
	5% WB	MCWBR	9.4	10.0	9.9	10.4	10.2	9.8	9.7	10.5	10.7	10.2	10.0	9.2		
		MCWBR	9.4	10.0	9.9	10.4	10.2	9.8	9.7	10.5	10.7	10.2	10.0	9.2		
(40) Clear-Sky Solar Irradiance	taub		0.235	0.256	0.290	0.324	0.337	0.344	0.344	0.338	0.312	0.289	0.262	0.234		
	taud		2.414	2.367	2.357	2.377	2.412	2.425	2.417	2.454	2.513	2.561	2.543	2.454		
	Ebn at Noon		280	294	298	295	294	291	290	287	287	279	268	268		
	Edh at Noon		21	27	32	36	36	35	35	32	28	23	18	18		
(44) All-Sky Solar Radiation	RadAvg		429	718	1078	1375	1540	1748	1728	1521	1144	766	445	354		
	RadStd		49	89	130	189	168	166	157	157	128	88	65	49		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (103 neighbors)	N/A	N/A	N/A	+1.15	+0.81	+0.72	N/A	N/A	+95	+23

Nomenclature: See separate page